

and all other unwanted devices and services if you are not using any of them. I strongly recommend that you study the basics about the services before disabling them, as it might affect the machine or operating system. Figure 6 shows the GUI of BUM.

Editing the rc file: To edit the rc file, you need to go to the rc directory. This can be done using the following command:

```
cd /etc/init.d.
```

However, root privileges are needed to access *init.d*, which basically contains start/stop scripts that are used to control (start, stop, reload, restart) the daemon while the system is running or during boot.

The rc file in *init.d* is called a run control script. During booting, *init* executes the rc script and plays its role. To improve the booting speed, we make changes to the rc file. Open the rc file (once you are in the *init.d* directory) using any file editor.

For example, by entering *vim rc*, you can change the value of *CONCURRENCY=none* to *CONCURRENCY=shell*. The latter allows certain startup scripts to be executed simultaneously, rather than serially.

In the latest versions of the kernel, the value should be changed to *CONCURRENCY=makefile*.

Figures 7 and 8 show the comparison of boot-up times before and after editing the rc file. The improvement in the boot-up speed can be noticed. The time to boot before editing the rc file was 50.98 seconds, whereas the time to boot after making the changes to the rc file is 23.85 seconds.

However, the above-mentioned changes don't work on operating systems later than the Ubuntu version 15.10, since the operating systems with the latest kernel use the *systemd* file and not the *init.d* file any more.

E4rat: E4rat stands for e4 'reduced access time' (ext4 file

```
The time after the unit is active or started is printed after the "@" character.
The time the unit takes to start is printed after the "+" character.

graphical.target @30.388s
├─multi-user.target @30.388s
│   └─kerneloops.service @19.552s +86ms
│       └─network-online.target @19.538s
│           └─NetworkManager-wait-online.service @15.328s +4.209s
│               └─NetworkManager.service @10.799s +4.522s
│                   └─dbus.service @9.162s
│                       └─basic.target @8.608s
│                           └─sockets.target @8.608s
│                               └─snapd.socket @8.473s +47ms
│                                   └─sysinit.target @8.469s
│                                       └─apparmor.service @2.885s +5.530s
│                                           └─local-fs.target @2.863s
│                                               └─run-user-121.mount @16.621s
│                                                   └─swap.target @2.998s
│                                                       └─swapfile.swap @2.713s +284ms
│                                                           └─systemd-remount-fs.service @2.231s +479ms
│                                                               └─systemd-journald.socket @2.178s
│                                                                   └─system.slice @2.178s
│                                                                       └─.slice @2.162s

lines 1-23
```

Figure 5: Output of *systemd-analyze critical-chain*

Activate	Description
☒	330bcb97ef14d708a3f5db273fe5cd73 kerneloops
☒	9726217383c5d7c9454fe7d0db64bda2 speech-dispatcher
☒	Fast remote file copy program rsync
☒	Configures your system for internet access via dns-clean
☒	Restores DNS pppd-dns
☐	40525278a400423d374c64c78bc211a3 friendly-recovery
☐	943f3288c1aaa379fca73a3ff1a35278 procps
☐	e009e7114cd0b15ac6dbe0d813ec9472 resolvconf
☐	11a29fdbf49a2e95dc09dd3531b75021 bootchart
☐	Manages your Internet connection networking
☐	Runs system housekeeping chores on specified anacron
☐	Discover services and hosts on a local network avahi-daemon
☐	.legacy-bootordering
☐	85ee5b58c8d635ba9041b52f81058494 console-setup
☐	c25fbd9bcc88ab95bcc33eea4228fd42

Figure 6: BUM

system only). It is a project developed by Andreas Rid and Gundolf Kiefer. E4rat is an application that helps in achieving a fast boot with the help of defragmentation. It also accelerates application startups. E4rat eliminates both seek times and rotational delays using physical file reallocation. This leads to a high disk transfer rate.

E4rat is available as a .deb package and you can download it from its official website <http://e4rat.sourceforge.net/>.

Ubuntu's default ureadahead package conflicts with e4rat. So a few packages have to be installed using the following command:

```
sudo dpkg purge ureadahead ubuntu-  
minimal
```

Now install the dependencies for e4rat using the following command:

```
sudo apt-get install libblkid1  
e2fslibs
```

Open the downloaded .deb file and install it. Boot data is now needed to be gathered properly to work with e4rat.

Follow the steps given below to get e4rat running properly and to increase the boot-up speed.

- Access the Grub menu while booting. This can be done by holding the shift button when the system is booting.
- Choose the option (kernel version) that is normally used to boot and press 'e'.
- Look for the line starting with *linux /boot/vmlinuz* and add the following code at the end of the line (hit space after the last letter of the sentence):

```
- init=/sbin/e4rat-collect or try -  
quiet splash vt.handsoff =7 init=/sbin/  
e4rat-collect
```

- Now press *Ctrl+x* to continue booting. This lets e4rat collect data after booting. Work on the machine, open and close applications for the next two minutes.